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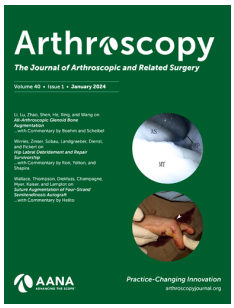

Injectable therapeutic peptides–an adjunct to regenerative medicine and sports performance?

Mikalyn T. DeFoor, MD, Travis J. Dekker, MD

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Mikalyn T DeFoor, MD[a] and Travis J Dekker, MD[b]

[a]San Antonio Military Medical Center, Fort Sam Houston, TX USA

[b]10th Medical Group, US Air Force Academy, CO, USA

Corresponding Author:

Travis J Dekker, MD

10th Medical Group, US Air Force Academy, 4102 Pinion Drive, CO, 80840 USA

Email: travisdekker88@gmail.com

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Injectable therapeutic peptides–an adjunct to regenerative medicine and sports performance?

Abstract

High level athletes and bodybuilders are constantly seeking novel therapies to enhance recovery and expedite return from injury—injectable peptides are a new and trending therapy that may be the wave of the future of in the realm of regenerative medicine research in treating joint injuries and osteoarthritis. Very early *in vivo* research of the pharmacokinetics suggests the possibility of Body Protection Compound 157 (BPC-157) appears to be at the forefront of therapeutic peptides with early demonstration of these experimental peptides optimizing endurance training, metabolism, recovery and tissue repair. While uUnregulated, yetand readily available for purchase over the internet, there is scarce orthopaedic literature investigating the clinical use and outcomes of these therapeutic peptides in tendon, muscle and cartilage injury. However, this has not slowed the recent exponential growth of the multi-billion-dollar industry in development of therapeutic peptides. As orthopaedic surgeons and team physicians, we should stay up-to-date with the latest pharmacokinetic, safety, ethical, and legal profilesregulations regarding synthetic peptide supplementation for injury recovery and sports performance optimization from our elite athletes to our fitness fanatics as these patients will continue to seek the latest and greatest in treatment options and will be approaching us with questions on their results, risks and benefits..

Letter To the Editor

A forthcoming topic of research in regenerative medicine and sport enhancement is concerning peptide therapy, which are naturally occurring and selective signaling molecules that bind to specific cell surface receptors to trigger an intracellular cascade of events.[1] Although a robust knowledge base of the application of therapeutic peptides is lacking in sports performance, early research, mostly *in vitro* and in animal models, suggests promising applications in endurance training and recovery through attenuating markers of muscle damage, stimulating robust anti-inflammatory responses, upregulating effects of

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skeletal muscle protein synthesis, and promoting collagen synthesis in tendons and extracellular matrix-protein synthesis in injured chondrocytes.[2,3]

There are several compounds investigated throughout the literature listed as therapeutic peptides affecting tissue and muscle repair (**Table 1**). Specifically for sports performance, Body Protection Compound 157 (BPC-157), a synthetic gastric-derived peptide, has gained significant attention with the possibility of optimizing sports performance, metabolism, recovery, and tissue repair—although the exact mechanisms need further explanation in human models.[3-5] These small peptides and peptide hormones may have a role for treatment across the full spectrum of sports medicine—from injury prevention, including the ability to change body composition and enhance endurance performance, to recovery and rehabilitation after injury through tendon, muscle, and cartilage repair.[4,5]

Orthobiologics, which gained significant popularity in sports medicine to reduce pain and expedite return to play in certain musculoskeletal injuries, include isolated growth factors, platelet-rich plasma (PRP), other autologous blood formulations, and cell therapy utilizing autologous cells derived from bone marrow, amniotic sources or adipose tissue.[6] Comparatively peptides (<40 amino acid chain length) differ from orthobiologics based primarily on size. They are small molecules derived from endogenous compounds which are generally tolerable in patients with a side effect profile including local skin reaction, decreased insulin sensitivity and/or water retention.[7,8] Both peptides and orthobiologics are highly specific to their target and can produce long-lasting effects.[8] However, peptides have the added benefit of synthetic production due to their small size, unlike orthobiologics which are extracted from living cells. Additionally, peptides demonstrate higher oral bioavailability, lower immunogenicity, longer shelf life, and stability at room temperature compared to orthobiologics.[9]There is a paucity of evidence in the current orthopaedic literature, but early pharmacokinetic and drug delivery studies in metabolic and oncologic disease have demonstrated increased use of peptide therapy through oral, transdermal, and injectable routes.[8] Only two studies in the current orthopaedic literature investigate therapeutic peptides in sports medicine, although with significant methodological limitations to draw concrete conclusions on the

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effectiveness of therapeutic peptides. Dressler et al conducted a randomized, case control study comparing oral supplementation of specific collagen peptide supplementation (SCP) versus placebo in the ability to decrease recurrent ankle instability.[10] While the SCP cohort revealed improved ankle stability and a smaller number of ankle injuries at 6 months, the follow-up data is limited by subjective outcomes without any discussion of the preparation, concentration or dosing of the oral supplementation. Lee and Padgett reviewed a case series of 17 patients who underwent intra-articular injections of BPC-157 for knee pain.[11] While over 90% of patients demonstrated improvement in knee pain at minimum 6-month follow-up, the majority of patients were treated for ligamentous and tendinous sprains, which often heal overtime without intervention. Additionally, this study did not measure improvement in function, quality of life or stiffness, and was conducted in the primary care setting.

Therapeutic peptides are widely available for purchase over the internet as dietary supplements and are here to stay with an estimated $35 billionvalue industry as of 2015, with a projected growth of 9% annually from 2016 to 2024.[8,12] As of 2019, there has been exponential growth in development with 60 US Food and Drug Administration (FDA)-approved peptide medications on the market, 140 peptide drugs in clinical trials, and more than 500 therapeutic peptides in preclinical development stages.[4,12] Specific to sports medicine and human performance, the potential uses of injectable peptide therapy are manifold but still begs the question—are they effective? BPC-157 is naturally isolated from human gastric juice, stable at room temperature, excreted in the urine and rapidly metabolized *in vivo* into single amino acids, making this pharmacokinetic profile favorable for translation into clinical practice.[9,12,13] However, further knowledge is required on the standard preparation, concentration and biomolecular properties before answering the question of effectiveness in human performance. However, we should be cautiously optimistic as early literature demonstrates a pharmacokinetic profile that has the ability to selectively target specific cell surface receptors for muscle and chondrocyte recovery with a limited side effect profile.

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Although a novelty in therapeutic interventions, BPC-157 is currently prohibited by the World Anti-Doping Agency (WADA) and, as of 2022, listed as a prohibited substance by all NCAA and professional anti-doping sporting agencies.[13,14] While BPC-157 is banned due to dosing concerns and lack of robust clinical trials, testing is limited in efficacy as this compound has a half-life of <30 minutes.[14] As therapeutic peptides become more mainstream, it is our obligation to counsel our patients and athletes appropriately. Currently, we do not recommend the use of BPC-157 in sports performance and recovery, as there are no randomized control trials investigating use in human subjects, and the science is clearly lacking to determine the overall effectiveness, safety profile and clinical indications for sports enhancement in athletes. However, with further research therapeutic peptides may offer a synthetic alternative or adjunct to regenerative medicine with a selective pharmacokinetic profile to aid muscle recovery and endurance training.

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Table 1. Select bioactive peptides used in sports performance **Synthetic Peptide**	**Description**	**Therapeutic Effects**
Collagen peptide supplementation (COL)[15]	Synthetic extra-cellular protein with high arginine and glycine composition	Increases contractile muscle tissue, stimulates connective tissue elasticity, and increases myofibrillar contractility
Body Protection Compound 157 (BPC-157)[5]	Gastric pentadecapeptide, 15 amino acid fragment (Gly-Glu-Pro-Pro-Pro-Gly-Lys-Pro-Ala-Asp-Asp-Ala-Gly-Leu-Val)	Enhances fibroblast activity, potent anti-inflammatory effects, stimulates collagen synthesis, and upregulates skeletal muscle protein synthesis
TB-500 (Thymosin Beta-4)[16]	Synthetic fraction of the protein 43 amino acid Thymosin Beta-4	Promotes differentiation of progenitor cells, potent anti-inflammatory effects, and induces angiogenesis

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Copper-binding peptide GHK-Cu)[16]	Copper binding peptide, glycyl-l-histidyl-l-lysine that functions as a complex with copper 2+	Stimulates function of dermal fibroblasts, modulates activity of metalloproteinases, and initiates collagen synthesis and breakdown
Ibutamoren (MK 677)[19]	Synthetic peptides that stimulate growth hormone-releasing hormone (GHRH) release from the hypothalamus	Stimulates secretion of growth hormone (GH) and Insulin-like Growth Factor 1 (IGF-1), increases protein synthesis for muscle mass, and increases metabolic rate to increase fat loss
CJC 1295[19]	Synthetic analogue of GHRH, stimulates release of GH from the pituitary gland	Extends half-life and bioavailability of GHRH, stimulates IGF-1 secretion, increases protein synthesis for muscle mass, and increases metabolic rate to increase fat loss

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